

Chapter 12: Congruent Figures

Worksheet 12: Congruent Figures

Learning Objective: Understand congruence of line segments, angles and triangles, and apply the SSS, SAS, ASA and RHS criteria to test triangle congruence.

Worked Example:

In triangles ABC and PQR, $AB=PQ$, $BC=QR$, and $AC=PR$. Are the triangles congruent? By which criterion?

Solution: Yes, by the SSS (Side-Side-Side) criterion, since all three corresponding sides are equal.

Questions (1–20)

1. What does “congruent” mean for two figures?
2. Name the four criteria used to prove triangle congruence.
3. If two line segments have the same length, are they congruent?
4. In congruent triangles ABC and DEF, if angle A = 50° , what is angle D?
5. What does SAS stand for?
6. In triangles ABC and XYZ, $AB=XY$, angle B = angle Y, $BC=YZ$. Which criterion proves congruence?

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7. In right triangles ABC and DEF, angle B = angle E = 90° , AC=DF (hypotenuse), and AB=DE. Which criterion applies?
8. Two triangles have all three corresponding angles equal but different side lengths. Are they congruent? Explain.
9. In triangles PQR and LMN, angle P = angle L, PQ=LM, angle Q = angle M. Which criterion applies?
10. If triangle ABC is congruent to triangle DEF, and AB= 7 cm, what is DE?
11. Two triangles ABC and PQR have AB=PQ= 6 cm, BC=QR= 8 cm, and angle B = angle Q = 60° . Are they congruent? State the criterion.
12. In triangle ABC, angle A= 40° , angle B= 70° . In triangle DEF, angle D= 40° , angle E= 70° , and AB=DE. Are the triangles congruent? By which criterion?

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13. A carpenter cuts two wooden triangular pieces with sides 5 cm, 6 cm, 7 cm and 5 cm, 6 cm, 7 cm. Are they congruent? Explain.
14. In triangles ABC and XYZ, $AB=XY$, $AC=XZ$, but angle A $\neq$ angle X. Can we conclude the triangles are congruent? Explain why or why not.
15. State the criterion used to prove congruence of two right triangles where one leg and the hypotenuse are equal in both.
16. Two triangular flags have sides 30 cm, 40 cm, 50 cm and 30 cm, 40 cm, 50 cm, but the second is a mirror image of the first. Are they still congruent? Explain.
17. In isosceles triangle ABC with $AB=AC$, a line AD is drawn from A to the midpoint D of BC. Explain why triangle ABD is congruent to triangle ACD.
18. A bridge truss uses congruent triangular sections. If one section has sides 4 m, 4 m, 5 m, describe how you would check another section is congruent to it.

19. Find two identical (congruent) objects at home or school, like two same-sized tiles. Explain how you know they are congruent.

20. Cut out two paper triangles with the same side lengths and check if they fit exactly on top of each other. Describe your observation.

Reflection Question: Why do you think engineers and architects rely on congruent triangles when building bridges and roofs? Write your idea.

Real-Life Application Question: Look for two congruent shapes in tiled flooring, a fence design, or window grills at home. Describe what makes them congruent.